A Bounded Remainder in the Sharp Logarithmic Causal Entropy Law

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Abstract

For finitely many centered finite increment laws of common positive variance, the upper Bellman semigroup has uniformly bounded integrated amplification on sufficiently broad Gaussian densities. A Gaussian telescoping argument proves this by induction, using a summable $t^{-3/2}$ defect. Combined with the existing pointwise residual bound for the logarithmic corrector, it yields an $O(1)$ remainder in both the mean and the unscaled W_1 Gaussian approximation. Consequently, for a stationary kernel of common entropy h and common information variance $v > 0$, the optimal causal seed entropy is $nh + (\kappa_+ - \kappa_-)(3v\sqrt{2\pi e})^{-1} \ln n + O(1)$.

Verification status. The new Gaussian amplification argument below is an analytic proof, not a Lean-checked extension. Its input from the existing development is the explicit corrector and pointwise residual majorant, not the conclusion that its error propagates boundedly. Historical priority has not been established. A convergent constant remainder is not claimed.

1 Setting and main result

Let $X_1, \dots, X_d$ have finitely supported laws satisfying

$$\mathbb{E}X_i = 0, \qquad \mathbb{E}X_i^2 = v > 0.$$

Set $\kappa_i = \mathbb{E}X_i^3$, $\kappa_- = \min_i \kappa_i$, $\kappa_+ = \max_i \kappa_i$, and

$$A_i f(x) = \mathbb{E}f(x - X_i), \qquad Tf = \min_i A_i f, \qquad Uf = \max_i A_i f.$$

Both operators preserve order and commute with translations. The operator U is positively homogeneous and subadditive, and

$$|Tf - Tg| \le U|f - g|, \qquad |Uf - Ug| \le U|f - g|. \tag{1}$$

Let $H = \mathbf{1}_{[0,\infty)}$, $G_n = T^n H$, and let ν_n be the probability law with CDF G_n. The mean of this envelope law is $e_n = \int_{\mathbb{R}}(H - G_n)$. The family is fixed throughout; constants are permitted to depend on it. Write ϕ, Φ for the standard Gaussian density and CDF.

Theorem 1. *With $c = (\kappa_+ - \kappa_-)/(3v\sqrt{2\pi e})$, there is a finite constant K such that for all integers $n \ge 1$,*

$$\boxed{W_1(\nu_n, N(c \ln n, vn)) \le K, \qquad |e_n - c \ln n| \le K.} \tag{2}$$

No lattice or aperiodicity assumption is needed.

Corollary 2. *For any finite stationary kernel with common action entropy h and common positive information variance v, in bits,*

$$\boxed{C_n = nh + \frac{\kappa_+ - \kappa_-}{3v\sqrt{2\pi e}} \ln n + O(1),}$$

where $\kappa_i = \mathbb{E}(-\log_2 P_i(Y_i) - h)^3$. The same expansion holds for the envelope mean and for exact seeds of entropy within a bounded additive amount of that mean.

Proof of the corollary from Theorem 1. The centered transcript-information envelope is G_n, and $L_n = nh + e_n$. Use the finite-tree sandwich $L_n \le C_n \le L_n + c_\star$ from [1]. □

2 Uniform Gaussian amplification

For $t > 0$ denote the $N(0, vt)$ density by

$$\gamma_t(x) = \frac{1}{\sqrt{2\pi vt}} e^{-x^2/(2vt)}.$$

Let P be convolution with $N(0, v)$, so $P\gamma_t = \gamma_{t+1}$. This is a comparison operator, not an additional available action.

Lemma 3 (One-step Gaussian defect)**.** *There is $D < \infty$ such that for all $t \ge 1$ and $x \in \mathbb{R}$,*

$$|U\gamma_t(x) - \gamma_{t+1}(x)| \le Dt^{-3/2}\gamma_{8t}(x). \tag{3}$$

Proof. Let $M = \max_i \max\{|y| : y \in \operatorname{supp} X_i\}$. Centered Taylor expansion and equality of the second moments give

$$\left| A_i\gamma_t - \gamma_t - \frac{v}{2}\gamma_t'' \right|(x) \le \frac{\mathbb{E}|X_i|^3}{6} \sup_{|a|\le M} |\gamma_t'''(x-a)|.$$

Also $\partial_t\gamma_t = (v/2)\gamma_t''$, whence

$$\left| \gamma_{t+1} - \gamma_t - \frac{v}{2}\gamma_t'' \right|(x) \le \frac{1}{2} \sup_{t\le s\le t+1} |\partial_s^2\gamma_s(x)|.$$

For finite constants D_3, D_4, uniformly in $t \ge 1$,

$$\sup_{|a|\le M} |\gamma_t'''(x-a)| \le D_3 t^{-3/2}\gamma_{8t}(x), \tag{4}$$

$$\sup_{t\le s\le t+1} |\partial_s^2\gamma_s(x)| \le D_4 t^{-2}\gamma_{8t}(x). \tag{5}$$

Here is a direct justification of uniformity in x. With $z = x/\sqrt{vt}$, $b = a/\sqrt{vt}$, one has $|b| \le M/\sqrt{v}$ and $(z-b)^2 \ge z^2/2 - b^2$. Gaussian spatial derivatives are a polynomial times their Gaussian density. Thus the ratio in (4), after removing $t^{-3/2}$, is bounded by a constant times $(1+|z|)^3 e^{-3z^2/16}$. For (5), use $\partial_s^2\gamma_s = (v^2/4)\gamma_s''''$ and $1 \le s/t \le 2$; the analogous ratio is bounded by a constant times $(1+|z|)^4 e^{-3z^2/16}$. Both suprema are finite. Finally $t^{-2} \le t^{-3/2}$ and $|\max_i a_i - b| \le \max_i |a_i - b|$, proving (3). □

Lemma 4 (Bounded integrated amplification)**.** *Choose $A \ge 1$ with $3D/\sqrt{A} \le 1/2$, where D is from Lemma 3. Then for every $t \ge A$, every $m \in \mathbb{R}$ and every integer $n \ge 0$,*

$$\boxed{\int_{\mathbb{R}} U^n[\gamma_t(\cdot - m)](x)\, dx \le 2.} \tag{6}$$

For example $A = \max\{1, 36D^2\}$ is admissible.

Proof. Translation invariance reduces the proof to $m = 0$. Positivity, subadditivity, (1), and $P^r\gamma_t = \gamma_{t+r}$ give the pointwise telescoping bound

$$U^n\gamma_t \le \gamma_{t+n} + D\sum_{r=0}^{n-1}(t+r)^{-3/2}U^{n-1-r}\gamma_{8(t+r)}. \tag{7}$$

For clarity, obtain this by comparing $U^n\gamma_t$ with $U^{n-1}\gamma_{t+1}$, then $U^{n-2}\gamma_{t+2}$, and so forth; each difference is bounded by the appropriate iterate of U applied to $|U\gamma_{t+r} - \gamma_{t+r+1}|$.

Use strong induction on n, simultaneously for every real $t \ge A$. The assertion at $n = 0$ is the Gaussian integral, equal to one. In (7), every remaining horizon is less than n and every Gaussian time $8(t + r)$ is at least A. The induction hypothesis therefore bounds its integral by two. Since

$$\sum_{r=0}^{\infty}(t+r)^{-3/2} \le t^{-3/2} + \int_0^{\infty}(t+s)^{-3/2}\,ds \le \frac{3}{\sqrt{t}},$$

the integrated right side of (7) is at most $1 + 6D/\sqrt{t} \le 2$. This closes the induction. There is no assumption that the supremum over all horizons is already finite. □

A general stability consequence. If $|f - g| \le b\gamma_t(\cdot - m)$ with $b \ge 0$ and $t \ge A$, then

$$\|T^n f - T^n g\|_1 \le 2b \qquad (n \ge 0). \tag{8}$$

This follows from (1) and Lemma 4. The same holds for sums of such majorants, with $2\sum b$ on the right. A further consequence, for $f \ge 0$ integrable, is $\int U^n(\gamma_t * f) \le 2\|f\|_1$ when $t \ge A$: take the supremum over policies inside the nonnegative integral against f, and apply Tonelli and (6).

3 The existing corrector input

We now use the following explicit analytic objects and bounds from the logarithmic-coefficient proof [2, 1]. Define

$$f(z) = \min_i\left[-\frac{\kappa_i}{6v^{3/2}}(z^2-1)\phi(z)\right], \quad a = \int f = -\frac{c}{\sqrt{v}},$$

$$J(z) = \int_{-\infty}^{z} 2(a\phi(y) - f(y))\,dy, \qquad B(z) = \phi(z)\int_0^z \frac{J(y)}{\phi(y)}\,dy.$$

The corrector solves $B'' + zB' + B = 2(a\phi - f)$, is integrable, and is bounded in absolute value by a polynomial times ϕ. For $t \ge 1$, set

$$z_t(x) = \frac{x - c\ln t}{\sqrt{vt}}, \qquad u_t(x) = \Phi(z_t(x)) + t^{-1/2}B(z_t(x)).$$

The established pointwise residual bound is

$$|u_{t+1}(x) - Tu_t(x)| \le K_0 t^{-2}(1 + z_t(x)^2)^4 e^{-z_t(x)^2/8}. \tag{9}$$

In the repository namespace `CausalSpectrum.LogGauss`, this is `residual_pointwise`, and `residual_majorant` packages it with its Gaussian window. These statements assume only centering, common variance $v > 0$, and the finite increment family. We do not assume any bounded propagation statement for these residuals.

Let $D_E = 64^4 e^{-63/16}$. Maximizing $(1 + z^2)^4 e^{-z^2/16}$ shows it is at most D_E. Therefore (9) implies, with $b_0 = K_0 D_E\sqrt{16\pi v}$,

$$|u_{t+1} - Tu_t|(x) \le b_0 t^{-3/2}\gamma_{8t}(x - c\ln t). \tag{10}$$

The right side has summable mass, and Lemma 4 prevents that mass from growing with the remaining horizon.

4 Proof of the bounded remainder

Fix any real $T \geq \max\{A, 2\}$ once and for all, independently of n. The initial error has a Gaussian majorant

$$|H(x) - u_T(x)| \leq b_1 \gamma_{s_0}(x - c \ln T) \tag{11}$$

for some finite b_1 and some $s_0 \geq A$. To verify this, choose $s_0 = 8T$. On each sufficiently distant tail, $H - \Phi(z_T)$ is a Gaussian tail and $B(z_T)$ is bounded by a polynomial times $\phi(z_T)$. Dividing by the broader Gaussian $\gamma_{8T}(x - c \ln T)$ gives a bounded ratio tending to zero. On the remaining compact interval the numerator is bounded and the denominator has a positive minimum. The jump of H at zero does not affect this pointwise bound.

Using (1) repeatedly, followed by (6), (10), and (11), gives

$$\begin{aligned} \|G_n - u_{T+n}\|_1 &\leq \int U^n |H - u_T| + \sum_{j=0}^{n-1} \int U^{n-1-j} |u_{T+j+1} - T u_{T+j}| \\ &\leq 2b_1 + 2b_0 \sum_{j=0}^{n-1} (T + j)^{-3/2} \\ &\leq 2b_1 + \frac{6b_0}{\sqrt{T}} =: K_1. \end{aligned} \tag{12}$$

This is uniform in n. Moreover

$$\|u_t - \Phi(z_t)\|_1 = \sqrt{v}\, \|B\|_1 .$$

Finally the Gaussians with means $c \ln(n + T)$ and $c \ln n$, and variances $v(n + T)$ and vn, can be coupled using the same standard normal variable. Their W_1 distance is at most

$$|c| \ln(1 + T/n) + \sqrt{2v/\pi}\, (\sqrt{n + T} - \sqrt{n}),$$

which tends to zero and is bounded for $n \geq 1$. The one-dimensional formula $W_1(\mu, \nu) = \int |F_\mu - F_\nu|$ and (12) prove the first part of (2). The difference of means is bounded by W_1, proving the second. This proves Theorem 1. □

5 Concrete consequence and scope

For the dyadic pair

$$P = (1/4, \underbrace{1/16, \ldots, 1/16}_{12}), \qquad Q = (\underbrace{1/8, \ldots, 1/8}_{6}, \underbrace{1/32, \ldots, 1/32}_{8}),$$

one has $h = 7/2$, $v = 3/4$, and $\kappa_\pm = \pm 3/4$. Dyadic exactness additionally gives $C_n = L_n$ at every horizon. Consequently

$$\boxed{C_n = \frac{7}{2} n + \frac{2}{3\sqrt{2\pi e}} \ln n + O(1).}$$

The proof concerns the infimum over all exact causal seeds. It does not restrict seeds to a particular construction.

The new argument does not use the logarithmic envelope-width bound; it controls the positive residuals directly. The broad Gaussian time in Lemma 4 is an auxiliary smoothing parameter, not a change to the original kernel. All constants may deteriorate as the kernel or its variance varies. We do not prove that $C_n - nh - c \ln n$ converges, identify a constant term, or treat unequal action entropies here. The value of G_* is unaffected.

References

[1] J. Mikulík, *Finite Shannon theorem*, https://github.com/jendamikulik/finite-Shannon-theorem, retrieved 12 October 2026. Inputs: `CausalSeed/LogCorrector.lean`, `LogResidual.lean`, `LogAmp.lean`, and the finite entropy sandwich. The bounded remainder proved here is a new analytic extension, not asserted to be present in those files.

[2] J. Mikulík, *The Sharp Logarithmic Coefficient of a Controlled Information Envelope*, research manuscript, 12 October 2026, available in the same repository.